



Iapetus' geophysics: Rotation rate, shape, and equatorial ridge

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Abstract

Iapetus has preserved evidence that constrains the modeling of its geophysical history from the time of its accretion until now. The evidence is (a) its present 79.33-day rotation or spin rate, (b) its shape that corresponds to the equilibrium figure for a hydrostatic body rotating with a period of ~ 16 h, and (c) its high, equatorial ridge, which is unique in the Solar System. This paper reports the results of an investigation into the coupling between Iapetus' thermal and orbital evolution for a wide range of conditions including the spatial distributions with time of composition, porosity, short-lived radioactive isotopes (SLRI), and temperature. The thermal model uses conductive heat transfer with temperature-dependent conductivity. Only models with a thick lithosphere and an interior viscosity in the range of about the water ice melting point can explain the observed shape. Short-lived radioactive isotopes provide the heat needed to decrease porosity in Iapetus' early history. This increases thermal conductivity and allows the development of the strong lithosphere that is required to preserve the 16-h rotational shape and the high vertical relief of the topography. Long-lived radioactive isotopes and SLRI raise internal temperatures high enough that significant tidal dissipation can start, and despin Iapetus to synchronous rotation. This occurred several hundred million years after Iapetus formed. The models also constrain the time when Iapetus formed because the successful models are critically dependent upon having just the right amount of heat added by SLRI decay in this early period. The amount of heat available from short-lived radioactivity is not a free parameter but is fixed by the time when Iapetus accreted, by the canonical concentration of ^{26}Al , and, to a lesser extent, by the concentration of ^{60}Fe . The needed amount of heat is available only if Iapetus accreted between 2.5 and 5.0 Myr after the formation of the calcium aluminum inclusions as found in meteorites. Models with these features allow us to explain Iapetus' present synchronous rotation, its fossil 16-h shape, and the context within which the equatorial ridge arose.

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1. Introduction

Iapetus is the most distant, regular satellite of Saturn. Peale (1986) noted that Iapetus' synchronous rotation period of 79.33 days is unexpected considering its large distance from Saturn (i.e., an orbit with a 3.51×10^6 km semi-major axis, or ~ 60 saturnian radii, R_s). Recent measurements by Cassini of Iapetus' low density and disequilibrium shape have made this satellite's dynamical state even more anomalous.

We have revisited this anomaly using coupled thermal and dynamical-evolutionary modeling. We used recently determined parameters for material properties and Iapetus' characteristics (see Tables 1 and 2). The most recent Iapetus data were obtained by the Cassini mission during the close flyby of Iapetus on December 31, 2005.

Section 2 of this paper is a discussion of the three outstanding geophysical properties of Iapetus that constrain its origin: its rotation state, its shape, and the presence of the equatorial ridge.

Section 3 describes the model and assumptions. We discuss the main constraints, which are imposed by initial composition and structure. The effects of possible compositions are tested by

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Table 1
Iapetus' physical properties

Parameter (unit)	Value	Reference
Mean radius (km)	735.6 ± 3.0 km	Thomas et al. (2007)
Biaxial ellipsoid radii (km)	$747.4 \pm 3.1 \times 712.4 \pm 2.0$	Thomas et al. (2007)
Min–max radii (km)	35.0 ± 3.7	Thomas et al. (2007)
GM (km^3/s^2)	120.5117 ± 0.0173	Jacobson et al. (2006)
Density (kg/m^3)	1083 ± 13	Thomas et al. (2007)

Note. Gravitational constant: $G = 6.672(59 \pm 84) \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$.

Table 2
Iapetus' dynamical properties

Parameter (unit)	Value	Reference
Semi-major axis (km)	3.5613×10^6	Yoder (1995)
Semi-major axis (R_{Saturn})	59.09	Yoder (1995)
Orbital period (days)	79.330183	Yoder (1995)
Orbital rate (rad/s)	9.1670093×10^{-7}	Yoder (1995)
Rotation period (days)	Synchronous	Yoder (1995)
Rotation rate (rad/s)	9.1670093×10^{-7}	Yoder (1995)
Eccentricity	0.0283	Yoder (1995)
Inclination (degrees)	7.52	Yoder (1995)

Note. R_{Saturn} : Saturn's radius = 60,268 km (equatorial radius).

varying the starting conditions: differing amounts of radionuclides and a range of volatile content. In terms of structure, we find that porosity is very important and its evolution must be included in the models.

Section 4 discusses the results of the modeling. We find self-consistent models, which lead to Iapetus' present-day rotation and shape. All of these successful models require formation of Iapetus early and inclusion of significant amounts of short-lived radioactive isotopes (SLRI). The main difficulty in modeling Iapetus is a shortage of heat. Without including SLRI we were not able to find realistic models that despin and still have the correct shape. Since the amount of SLRI heat needed for a particular model maps directly to formation time, one obtains the time of accretion for that model.

The models do not explicitly tell us about the formation of the equatorial ridge. However, through the prediction of decreasing surface area on Iapetus during despinning to synchronous rotation they provide a possible rationale as to why the ridge formed. Tracking surface area versus time provides some times when the ridge may have been formed.

Section 5 is a discussion of the broader implications of the models. Included are modeling techniques, convection, ridge formation, comparison with other satellites, the origin of Iapetus, and absolute chronometry and its implications.

Iapetus is a very intriguing object. Due to its circumstances, it did not have enough heat to evolve as far as other satellites. In this sense, it only partially evolved. It is this state of “suspended animation” that provides a unique opportunity for geophysical investigations.

2. The data

Three features of Iapetus' current state provide critical constraints for our models, which in turn yield information about

Iapetus' past. These features are described in the next three sections.

2.1. Spin and orbit

Today, Iapetus' rotation period is 79.33 days and synchronous with its orbital period. Despinning to reach synchronicity is a result of tidal dissipation, a process that has occurred for all the regular jovian and saturnian satellites (except chaotic Hyperion). However Peale (1986) noted that Iapetus' synchronous spin was unexpected, because of its large semi-major axis ($D = 3.51 \times 10^6$ km, i.e., $\sim 60 R_s$) and the very strong (D^{-6}) dependence of the rate of despinning on distance to the planet.

Iapetus has a mean density of $1083 \pm 13 \text{ kg}/\text{m}^3$ (Thomas et al., 2007). This means that it has a radiogenic bearing rock mass fraction of $\sim 20\%$, assuming the body is not porous. Iapetus' mean density indicates that it is most likely composed of water ice and chondritic carbonaceous chondritic material (from the kronian subnebula) with an enrichment in volatiles and, possibly, light hydrocarbons (Johnson and Lunine, 2005). Its composition and current orbital state are fully consistent with formation in its present place (Ward, 1981) as part of the saturnian system. (This will be discussed further in Section 5.)

Iapetus is “dynamically frozen” (to use a phrase from McKinnon, 2002). Its eccentricity is 0.0283, which suggests little dynamical evolution. Its orbital inclination is ~ 7.49 degrees and its Laplace plane is inclined 14.968 degrees with respect to Saturn's equator.

2.2. Shape

Cassini images show that Iapetus is an oblate spheroid with a difference between its equatorial, a , and polar radii, c , of 35.0 ± 3.7 km (Thomas et al., 2007) (see Fig. 1). The equatorial radii a and b are the same to within 2 km, i.e., less than the uncertainty in the measurements. The residuals to limb fits over the wide range of longitudes and latitudes available, have an rms of 4.0 km (0.54% of the mean radius), and show, as does inspection of the images, that this shape is indeed an ellipsoid with superposed craterform topography. A symmetric difference of 34 km with such small residuals could not form by random cratering by large impacts. Nor is modification of a triaxial equilibrium form possible for two reasons. First, the difference of observed and predicted intermediate axes is over 26 km, a value not allowed by the measurement uncertainty. Second, to maintain an $(a - c)$ of 34.5 km with its slow rotation and observed mean density, Iapetus would have to be largely hollow with a high density, thin shell, clearly not physically plausible. The predicted $a - c$ for a homogeneous Iapetus is only ~ 10 m for hydrostatic equilibrium for the current spin period, as illustrated in Fig. 1. Thus, Iapetus has the largest non-hydrostatic anomaly known for a satellite larger than 1000 km in radius, with the Moon being a distant second (Garrick-Bethell et al., 2006).

If Iapetus' interior is homogeneous, the observed figure is only plausibly explained as the shape for a body in hydrostatic equilibrium with a rotation period of $\sim 16 \pm 1$ h. A differentiated

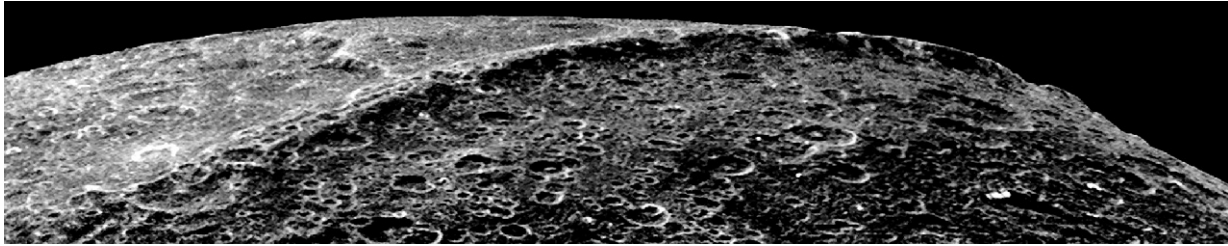


Fig. 2. Iapetus equatorial ridge as observed by Cassini ISS (Porco et al., 2005). Toward the left the ridge appears to have splayed into several approximately parallel components. The available images do not have enough detail to reveal how the ridge was formed.

tectonic (rather than a volcanic) event. We know little about the ridge's geological evolution.

If the ridge was created by folding, then its spatial wavelength of ~ 200 km implies that the lithospheric thickness was ~ 15 – 20 km, according to the relationship between the wavelength of folds and the thickness of the lithospheric shell that formed them (Turcotte and Schubert, 2002). Thus, the minimum lithospheric thickness required to preserve a 20-km high feature is about 15–20 km. However, the ridge could have been somewhat higher and subsequently relaxed to its present height. It is also possible that it resulted from faulting. Obviously, further imaging and interpretation are needed in order to better constrain the lithospheric thickness and the possible formation processes.

It should be emphasized that this feature is small compared to the global equatorial bulge. Removal of the ridge from limb data reduces the equatorial ellipsoid fit axes by only ~ 1 km. The volume of the ridge is less than 5% of the equatorial bulge, and is clearly a surimposed form. Although the formation of the ridge and bulge may be related (see below), the significance of the oblate spheroid shape is independent of the formation mechanism and morphology of the ridge.

2.4. Objectives of this paper

The present paper reports our effort to find a scenario or scenarios that can explain both the present shape and rotational state of Iapetus, while being compatible with the preservation of the equatorial ridge. This scenario is based on the assumption that the shape's oblateness is a fossil from a period when Iapetus was spinning more rapidly than present.

Solving this problem requires simultaneous modeling of the thermal and orbital evolution of the satellite. Indeed, despinning, shape, and lithospheric evolution are functions of the thermal structure of the satellite (dissipation factor and Love number k_2). Thermal events determine the rotational and geological events leading to the present condition. For the satellite to despin while preserving its shape, it is necessary for the interior to still be dissipative after a lithosphere thick enough to support the equatorial bulge has developed. This provides constraints on the internal evolution chronology and suggests a period during which the body was spinning rapidly and its global rigidity was low enough for hydrostatic equilibrium to be achieved. Despinning must continue as the lithosphere becomes thick enough to freeze its shape, i.e., warm temperatures are reached in the deep interior of the satellite. We believe that external heat sources (such as impact energy) are not important

since they would disrupt the overall ellipticity of the shape and lead to its relaxation, contrary to Iapetus' current state.

In the following sections we explain our models and test the possible scenarios meeting these requirements.

3. Internal evolution modeling

In this section we describe and justify the initial conditions used in the models. Then we explain the algorithms used for handling coupled thermal, dynamical, and geological processes.

3.1. Initial conditions

3.1.1. Composition

For the rock component we use an ordinary chondritic composition based on Wasson and Kalleyman's (1988) elemental abundances (Table 3). The corresponding density (used for the calculation of the rock-metal fraction) is 3510 kg/m^3 . Radionuclide data for the long-lived radioactive isotopes (LLRI) are gathered in Tables 4 and 5.

Iapetus' icy component is assumed to be primarily water on cosmochemical grounds (e.g., Lewis, 1971), but it is likely that it also contains various volatile "contaminants." Carbon dioxide has been detected in small quantities by the Cassini Visual Infrared Mass Spectrometer (VIMS) (Buratti et al., 2005), but its origin is not established. Based on models of the properties and evolution of satellite-forming disks around giant planets, it is possible that ammonia is present at amounts relative to water of up to 11 wt% (Mousis et al., 2002). Recent measurements by Cassini RADAR (Ostro et al., 2006) identify ammonia as a contaminant that could explain RADAR observations at different wavelengths. The various "contaminants" may affect geophysical models (through the depression of the ice melting point)

Table 3

Initial isotopic abundances for the compositional model, ordinary chondrite, considered in this study

Property (unit)	Value
Density (kg/m^3)	3510
^{26}Al (ppb)	600
^{60}Fe (ppb)	22.5–225
^{53}Mn (ppb)	25.7
^{40}K (ppb)	1104
^{232}Th (ppb)	53.8
^{235}U (ppb)	8.2
^{238}U (ppb)	26.2

by direct collapse can easily meet this time constraint (Mayer et al., 2002). The nucleated instability model of giant planet formation may be more relevant to the giant planets of our own Solar System, with their supersolar heavy element abundances and regular satellite systems, but only under certain restricted conditions do they seem able to produce gas giant planets within a few million years (Lissauer and Stevenson, 2007).

A somewhat weaker constraint on giant planet formation time scales comes from dynamical calculations that show the presence of Jupiter and Saturn to be important determinants of the final architecture of the terrestrial planet orbits and masses (O'Brien et al., 2006), coupled with the fact that much of the growth of the Earth was completed within 10–30 million years after CAIs based on the Hf–W isotopic system. Our time scale result provides an additional indication that giant planet formation was relatively rapid in our own Solar System and gives impetus to the further development and elaboration of the nucleated instability models for giant planet growth.

The precision of the Iapetus modeling result will help support a better understanding of the links between the different chronological scales such as those derived from nucleosynthesis, formation of the first solids, meteorites, crater counting, and radiometric dating.

6. Conclusions

We have modeled the geophysical history of Iapetus based upon discoveries from the *Cassini–Huygens* mission, modeling of several interconnected physical processes, and updated properties for materials and the processes that operate upon them. We have considered a range of different initial assumptions, i.e., the initial porosity distribution in the satellite, initial temperature profile at the end of accretion, the presence or absence of SLRI, the presence or absence of significant amounts of ammonia, and the date of accretion with respect to CAIs.

We show that despinning is a non-linear process, and that it has significant effects on Iapetus' geology. Depending upon the initial conditions, synchronous rotation is achieved between 200 Myr and 1 Byr after formation. At the time of despinning the lithosphere must have become strong enough to preserve the non-hydrostatic figure seen today. It is important to note the key role played by SLRI in decreasing the porosity early in Iapetus' history. This allows heat to escape faster and leads to the development over the long run of a thick, mechanically strong, lithosphere, suitable, just in time, for preserving the 16-h figure.

We found that there were two large reductions in surface area that took place in Iapetus' history. The first was due to a decrease in porosity and the second was due to despinning. We note the potential geological role of either or both of these two events in the formation Iapetus' equatorial ridge.

Models without SLRI do not work no matter what the ammonia concentration. With SLRI we have a range of models that work, although such models including ammonia require more study.

Another general result is that in all of the models where convection may be an issue, the onset of convection in the outer

layer acts to frustrate early despinning. Since Iapetus demonstrably did despin to synchronous rotation, we conclude that convection, if it occurred, did not prevent the formation of a low-viscosity core.

Given the scope and importance of the conclusions that have come out of this study, further modeling and investigation is definitely warranted. We have taken care throughout the paper to point out the various uncertainties, limits on our knowledge, and paths not pursued. All of these are worthy topics for future research.

The notion that a high precision, absolute chronology of the outer Solar System can be obtained through the properties of the natural satellites is truly novel.

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Appendix A. Thermal evolution

This appendix refers to Section 3.2.1 in the text.

A.1. Conduction equation

Heat is transferred by conduction using

$$\frac{\partial(k(T)\partial T(r)/\partial r)}{\partial r} + \frac{2}{r}\left(k(T)\frac{\partial T(r)}{\partial r}\right) = \rho(r)C_p(T)\left(\frac{dT(r)}{dt}\right) - H(r),$$

where T is temperature (in Kelvin), r local radius, k thermal conductivity, ρ material density, C_p specific heat, t time, H internal heating (e.g., radiogenic, tidal dissipation).

A.2. Accretional heat

Accretional heat is computed after Squyres et al. (1988). The temperature increase due to accretion is small. It can reach up to 90 K in the largest medium-sized satellites Rhea and Iapetus,